

Arma 3 Mod Automation Guide

APCA Server Automation - Feature Roadmap

This document outlines the strategic roadmap for expanding the APCA Server Automation toolkit for Arma 3 mod and server management.

Phase 1: Core Functionality (Complete)

These foundational features have been implemented and packaged:

- **Mod Downloader** (`download_mods.py`): Automates downloading from Steam using `mods.json`.
- **Sanitizer** (`tolower.sh`): Normalizes mod folder names for Linux.
- **LGSM Config Updater** (`update_lgsm_config.sh`): Automatically updates `arma3server.cfg` with the latest mods.
- **Swiftly Deployment** (`deploy_swifty.sh`): Pushes mods to HTTP for client-side syncing.
- **Documentation**: Markdown and PDF guides for server admins.

Phase 2: Quality of Life & Maintenance

Mod List Diff Tool

Compare `mods.json` with actual mods in `/cmods` and `/omods`. Identify:

- Missing mods
- Extra mods not listed
- Mods with mismatched folder names

Interactive CLI Menu

User-friendly terminal interface using `dialog` or `whiptail`:

- Add/remove mods from JSON
- Trigger download/deploy
- View logs or summary

Log Rotation and Summary

Automated log archiving with:

- Date-stamped archives
- Summary reports: success/failures, mod count, runtime

Mod Dependency Checker

Scan known mods (e.g., ACE, RHS) and ensure dependencies are also listed or installed.

Self-Check Script

Validates system state:

- Folder permissions
- Disk space
- Internet access
- SteamCMD authentication

Arma 3 Mod Automation Guide

Phase 3: User Integration

Discord Bot Integration

Use Discord to:

- Upload mod lists
- Trigger mod downloads and deployments
- View status updates and logs

Security: restrict to specific roles; log all actions.

Web Dashboard (Optional: Flask or Node.js)

A lightweight UI with:

- Upload field for HTML or JSON
- Mod list viewer
- Start/stop server buttons
- Logs view
- Auth via password or Discord OAuth

Phase 4: Advanced Features

Mod Version Tracker

Maintain a local version registry and periodically check Workshop for updates.

Auto-notify or schedule update/download based on flags.

Auto-Restart on Mod Update

Use `inotify` or polling to monitor mod folders for changes. If a mod is updated:

- Re-run sanitization and LGSM update
- Restart Arma server

Cloud-Hosted Swifty Sync

Push mods to AWS S3, R2, or another CDN so clients download faster.

Useful for global player bases.

Backup and Rollback

Keep compressed snapshots of:

- `mods.json`
- Installed mods
- LGSM configs

Allow reversion in case of error or corruption.

Phase 5: Community and Mod Metadata

Arma 3 Mod Automation Guide

Shared Mod Presets

Save and share `mods.json` files:

- Upload to GitHub or web portal
- Tag by mission/scenario

Mod Tagging and Ratings

Enhance `mods.json` structure:

```
```json
{
 "id": "123456",
 "name": "ACE",
 "type": "core",
 "tag": "required",
 "rating": 5
}
```

Display in CLI, Discord, or web UI.

---

## ## Implementation Recommendations

Phase	Features	Complexity	Impact
2	CLI menu, mod diff, log tools	Low	High
3	Discord bot, web UI	Medium	High
4	Auto updates, rollback, CDN sync	High	Medium
5	Community tools, rating/tagging	Medium	Medium

This roadmap offers a robust, modular plan to evolve APCA Server Automation from a command-line toolset into a comprehensive mod and server orchestration suite.